

Demonstration of Proposed Features

Date	1 July 2026
Team ID	N/A
Project Name	Rising Waters
Maximum Marks	1 Mark

The Rising Waters: A Machine Learning Approach to Flood Prediction system was successfully developed and demonstrated as a web-based application. The application allows users to enter rainfall-related parameters through an interactive interface and instantly predicts whether there is a Flood Chance or No Flood Chance using a trained Machine Learning model

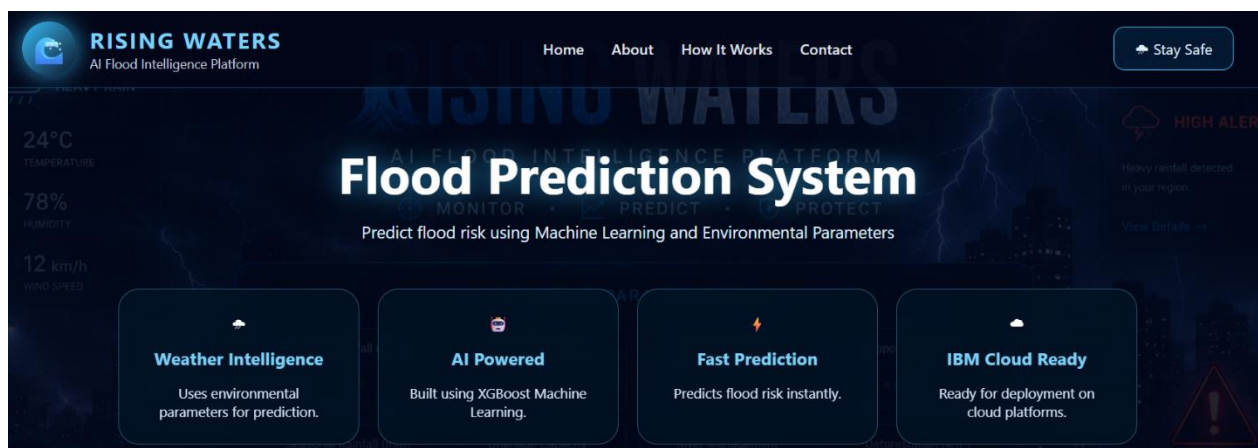
S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Rainfall Data Input	Allows users to enter annual and seasonal rainfall values through a web form.	Implemented	Yes	Successfully accepts user input.
2	Flood Prediction using Machine Learning	Predicts flood occurrence using the trained Random Forest model.	Implemented	Yes	Generates accurate Flood/No Flood predictions.
3	Interactive Web Interface	Responsive frontend developed using HTML, CSS, and JavaScript with rainfall animation.	Implemented	Yes	User friendly interface with Smooth navigation
4	Prediction Result Display	Displays Flood or No Flood result dynamically based on model prediction.	Implemented	Yes	Results are displayed instantly after prediction.

5	Cloud Deployment	Flask application deployed successfully on the Render cloud platform.	Implemented	Yes	Application is publicly accessible through the deployed URL.
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Feature Implementation Summary:

Item	Value
Total Features Proposed	5
Total Features Implemented	5
Total Features Demonstrated	5
Overall Implementation Rate (%)	100%

Attachments:



Enter Parameters

☔ Monsoon Intensity

0 - 10

🏞️ Terrain & Drainage

0 - 10

🌊 River Management

0 - 10

🌳 Deforestation

0 - 10

🏠 Urbanization

0 - 10

🌍 Climate Change

0 - 10

🏰 Dam Quality

0 - 10

🏗️ Siltation

0 - 10

🌾 Agricultural Practices

0 - 10

🏠 Encroachments

0 - 10

🚒 Disaster Preparedness

0 - 10

🔧 Drainage Systems

0 - 10

🌊 Coastal Vulnerability

0 - 10

⚠️ Landslide Risk

0 - 10

💧 Watersheds

0 - 10

🏢 Infrastructure

0 - 10

👤 Population Density

0 - 10

🌿 Wetland Loss

0 - 10

🏘️ Urban Planning

0 - 10

🏛️ Administrative Factors

0 - 10

🔮 Predict Flood Risk

✅ **NO FLOOD
RISK DETECTED**

Great news! Our **Machine Learning model** predicts **no significant flood risk** based on the rainfall conditions entered.

Flood Risk Level

%

Current flood risk is low. Continue monitoring weather updates.

🛡️ Stay Prepared

🌧️ Continue monitoring weather forecasts.

👁️ Observe rainfall conditions regularly.

📱 Stay connected with emergency alerts.

🚰 Keep drains and waterways clear.

🧰 Maintain an emergency preparedness kit.

🚗 Avoid unnecessary travel during heavy rain.